UGANDA ADVANCED CERTIFICATE OF EDUCATION
RESOURCE MOCK EXAMINATIONS – 2026
PHYSICS
Paper 2
(Practical)

SCORING GUIDE

ITEM 1

1. PLANNINFG FOR THE EXPERIMENT (PE)

Aim:

An investigation to assess the dimensional standards of the bare wire by determining its diameter, d. ✓

Or

An investigation to determine the diameter of the bare wire.

Variables:

- Independent variables – length of the bare wire. ✓
- Dependent variables – balance lengths measured along the metre bride wire. ✓
- Controlled variables – nature of the material of the bare wire, diameter of the bare wire. ✓

Hypothesis:

The diameter, d of the sample of the bare wire is in the range

$3.6X10^{-4}m$ to $3.9X10^{-4}m$. ✓

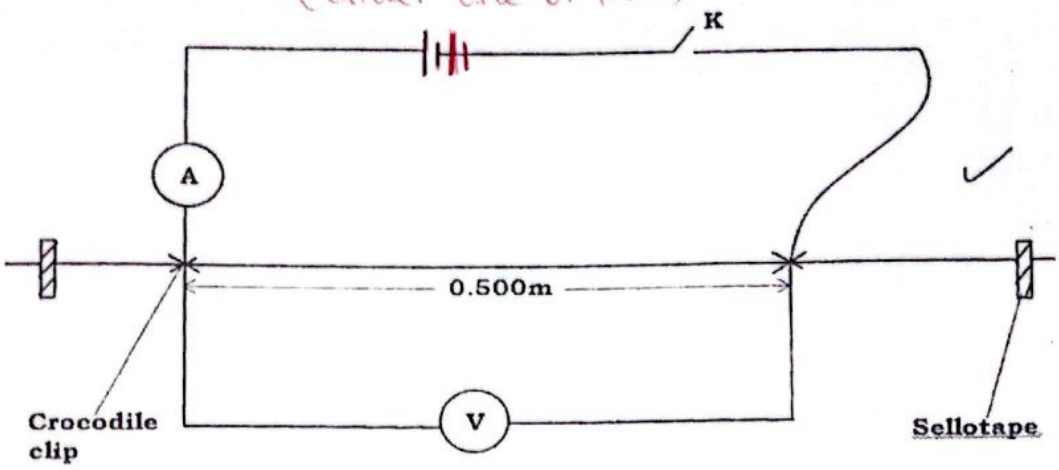
Key points = 5 key points

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Diagram | 01 Score
Closing | 01 Score
Quantity & Instrument | - Voltmeter, V - Current I, by Ammeter (04 Score

2. METHOD OF THE EXPERIMENT (ME)

Method 1 (Either one or two)



(a) The apparatus were arranged as shown in the set up above.

(b) The switch, K was closed.

(c) The potential difference across the 0.500m length of the sample bare wire was measured using a voltmeter and recorded as V (Quantity & Instrument)

(d) The electric current through the bare wire was measured using an ammeter and recorded as I. (Quantity & Instrument)

(e) The switch K was opened.

(f) The value of d_1 was determined from the expression: $d_1 = \sqrt{\frac{0.500X4\rho I}{\pi V}}$

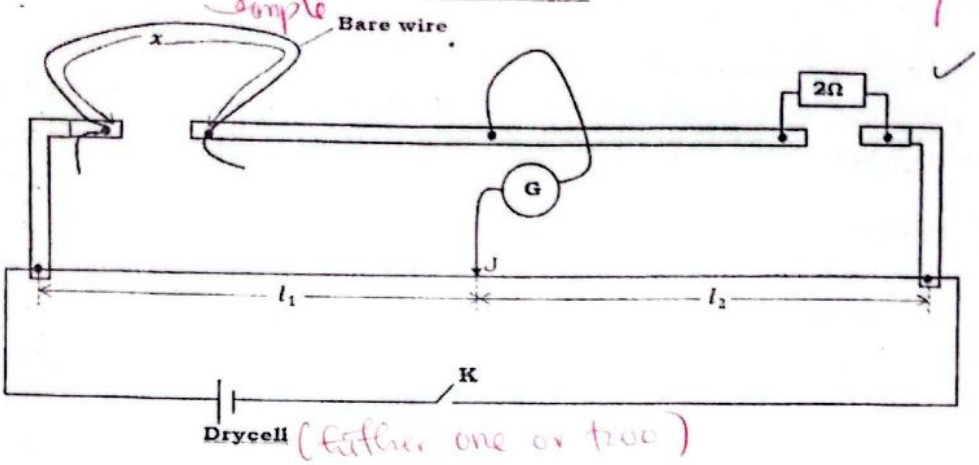
OR $d_1 = \sqrt{\frac{2\rho I}{\pi V}}$. Where $\rho = 4.9X10^{-7}\Omega m$

coherence

Key points = 9 key points

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Method 2



(either one or two)

(a) The apparatus were arranged as shown in the diagram above.

(b) The length, x of the bare wire was adjusted to x = 30.0cm.

(c) The switch K was closed.

(d) The sliding contact, J was moved along the metre bridge wire until the galvanometer, G showed no deflection.

(e) The balance lengths, l_1 and l_2 were measured using a metre rule and recorded. (Quantity (l_1 & l_2) & instrument)

(f) The switch K was opened.

(g) The steps (b) to (f) were repeated for other values of x = 40.0, 50.0, 60.0, 70.0 and 80.0cm.

(h) The results were tabulated including values of $R_x = 2\frac{l_1}{l_2}$

(i) A graph of R_x against x was plotted.

(j) The slope, S of the graph was determined.

(k) The value of d_2 was obtained from the expression: $d_2 = \sqrt{\frac{4\rho}{\pi S}}$

Where $\rho = 4.9X10^{-7}\Omega m$.

(l) The diameter, d of the sample of the bare wire was determined from the expression $d = \frac{d_1+d_2}{2}$

Coherence

14 Key Points

$R = \frac{\rho l}{A}$

But $A = \frac{\pi d^2}{4} \rightarrow R = \frac{4\rho l}{\pi d^2}$

$d_1 = \sqrt{\frac{4\rho l}{\pi R}}$

But $R = \frac{V}{I}$

$d_1 = \sqrt{\frac{4\rho l I}{\pi V}}$

$l = 0.50$

$d_2 = \sqrt{\frac{4\rho l}{\pi R}}$

But $\frac{l}{R} = \frac{1}{S}$

$d_2 = \sqrt{\frac{4\rho}{\pi S}}$

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Errors and precautions (Any one Correct Error with a precaution)

- Parallax errors during the measuring of the balance lengths/potential difference/current. This was mitigated/minimized by positioning the eye normally/perpendicularly the scale of the metre rule/voltmeter/ammeter. (1)
- Drainage of the dry cell. This was mitigated/minimized by keeping the switch open whenever the set up was not in use.

ME = (9 key points) + (15 key points) = 24 key points

3. DATA PRESENTATION (DP)

Method 1

V = 1.35V I = 0.62A (Quantity, Value & Unit)

Any Value.

Method 2

$x(cm)$	$x(m)$	$l_1(cm)$	$l_2(cm)$	$\frac{l_1}{l_2}$	$R_x(\Omega)$
30.0	0.300	34.6	65.4	0.53	1.06
40.0	0.400	40.6	59.4	0.68	1.36
50.0	0.500	44.4	55.6	0.80	1.60
60.0	0.600	50.6	49.4	1.02	2.04
70.0	0.700	54.7	45.3	1.21	2.42
80.0	0.800	61.2	38.8	1.58	3.16

Key points = 10 key points +1 = 11 Key points

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4. DATA ANALYSIS

Description	score	Key points
• Title of the graph: A graph of R_x against x	1	1
• Perpendicular axes drawn with arrows	1@	2
• Correctly labelled axes: VA: R_x (Ω) HA: x (m)	1@	2
• **Uniform**, convenient and suitable scales	1@	2
• Starting values sated/indicated	1@	2
• Five correctly plotted points.	1@	5
• Line of best fit drawn. (provided at least 3 points are correctly plotted)	1	1
• Slope indication: A right angled triangle drawn and touching the line of best fit.	1	1
• **Slope**		
- Coordinates correctly and accurately read to the number of d.ps of one small box on the respective axis.	1@	4
- Slope value correctly calculated and unit	1 (value) 1 (unit)	2
Total		**22 key points**

Key points = 22 key points

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5. CONCLUSION AND RECOMMENDATION

	Scores	Key points
• $d_1 = \sqrt{\frac{2\rho l}{\pi V}} = \sqrt{\frac{2X(4.9X10^{-7})X0.62}{\pi X1.35}} = \sqrt{\frac{6.1X10^{-7}}{4.24}}$ ✓ ✓ ✓ $= 3.8X10^{-4}m$ ✓	1 1(Value) 1 (unit)	1 2
• - $d_2 = \sqrt{\frac{4\rho}{\pi S}}$ - Substitution ✓ - Value and unit ✓ ✓	1 1 (value) 1 (unit)	1 2
• - $d = \frac{d_1+d_2}{2}$ - Substitution ✓ - Value and unit ✓	1 1 (value) 1 (unit	1 ~~2~~ 1
• The diameter of the sample of the bare wire is --- **(value stated)** which is in the range of $3.8X10^{-4}$ m to $3.9X10^{-4}$m ✓ • Therefore the bare wires satisfy specific dimensional standards. ✓	1 1	1 1
		~~11 key points~~ 10 key points

SUCCESS CRITERIA

BASIS CODE	Key points	LEVEL 4	3	2	1
PE	5		4 – 5	2 – 3	1
ME	24	18 – 24	12 – 17	6 – 11	1 – 5
DP	10	8 – 10	5 – 7	3 – 4	1 – 2
AN	22		15 – 22	8 – 14	1 – 7
CR	11			6 – 11	1 – 5

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ITEM 2

1. DATA PRESENTATION

- $W_0 = 0.75N$ $W_1 = 1.75N$

M(kg)	t(s)	T(s)	$T^2(s^2)$
0.100 0.200 0.300 0.400 0.500 0.600	Increasing trend	• 4 d.ps if largest value of $t < 20$ • 3 d.ps if largest value of $t \geq 20$	• 4 d.ps if largest value of $t < 20$ • 3 d.ps if largest value of $t \geq 20$

Key points = 9 key points

2. DATA ANALYSIS

Description	score	Key points
• Title of the graph: A graph of T^2 against M	1	1
• Perpendicular axes drawn with arrows	1@	2
• Correctly labelled axes: VA: $T^2(s^2)$ HA: M(kg)	1@	2
• **Uniform**, convenient and suitable scales	1@	2
• Starting values sated/indicated (Horizontal axis must start from zero)	1@	2
• Five correctly plotted points.	1@	5
• Line of best fit drawn. (provided at least 3 points are correctly plotted)	1	1
• Slope indication: A right angled triangle drawn and touching the line of best fit.	1	1
• **Slope**		
- Coordinates correctly and accurately ready to the number of d.ps of one small box on the respective axis.	1@	4 2
- Slope value correctly calculated and unit	1 (value) 1 (unit)	2
• Intercept correctly read to 2 or 3dp and unit	1 (value) 1 (unit)	2

Key points = 24 key points 17

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3. CONCLUSION AND RECOMMENDATION

		Key points
• $\alpha_1 = \frac{10000(W_1 - W_0)}{g}$ $= \frac{10000(1.75-0.75)}{9.81}$ ✓ (substitution) = **1019kgm^{-3}** ✓	1 1(Value) 1 (unit)	**1** **2**
• - $\alpha_2 = 10000\left(\frac{C}{S} - \frac{W_0}{g}\right)$ - Substitution ✓ - Value and unit ✓	1 1 (value) 1 (unit)	**1** **2**
• $\alpha = \frac{\alpha_1 + \alpha_2}{2}$ - Substitution ✓ - Value and unit ✓	1 1 (value) 1 (unit)	**1** **2**
• The density of the carbonated soda is **---(value stated)** which is in the range of $9.8X10^2$ kgm^{-3} to $1.1X10^2$kgm^{-3}. ✓ Feedback: • Therefore, the level of concentration of the carbonated soda is high ✓ **OR** The carbonated soda has a reasonable level of sweetness	1 1	**1** **1**
		8 key points 9 Key points

SUCCESS CRITERIA

		LEVEL			
	KEY POINTS	**4**	**3**	**2**	**1**
DP	**9**	7 - 9	5 – 6	3 – 4	1 - 2
AN	**24**		16 – 24	8 – 15	1 - 7
CR	**8**			4 – 8	1 - 3

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